

Bakare Olasubomi Numerical PDE Homework 1 Solution

Math 341, Fall 2026
Due, Sep. 9th, 2026.

1. **Exercise: 1** Given the initial value $p_0 = 0.5$, and plotting the solution error $|p_n - p|$, we see a linear convergence with asymptotic error ratio of 0.5, that is:

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} \approx \frac{1}{2} \quad (1)$$

We observe that numerical scheme to diverge from $n = 28$ due to floating point error. We have linear convergence because on a logarithmic plot we have that the error forms a straight line with a negative slope, meaning the error is multiplied by approximately $\frac{1}{2}$ at every iteration step rather than squaring as in standard quadratic Newton convergence.

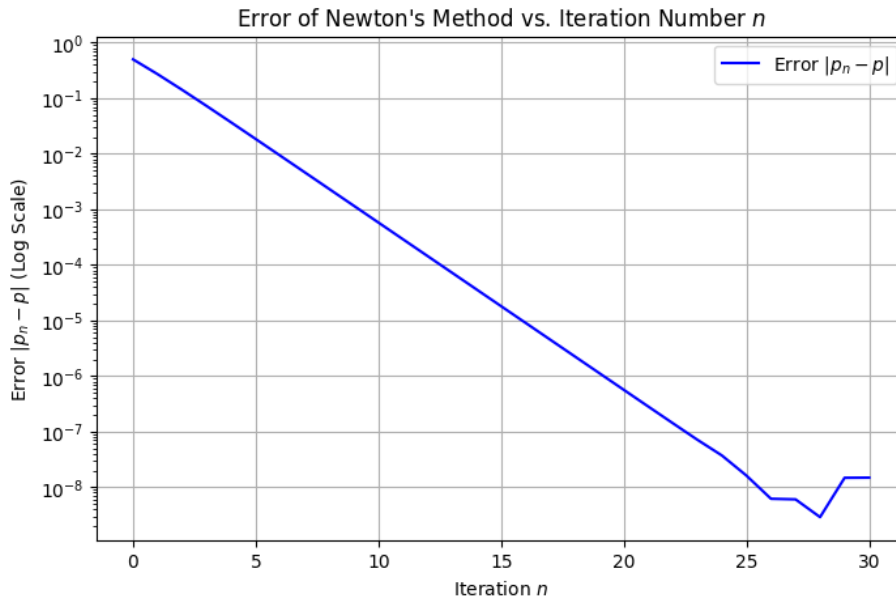


Figure 1: Newton Method

More explicitly we have that:

$$N_f(x) = x - \frac{f(x)}{f'(x)} \quad (2)$$

Differentiating $N_f(x)$;

$$N'_f(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = \frac{f(x)f''(x)}{(f'(x))^2} \quad (3)$$

Evaluating $N'_f(0)$.

$$f(x) = x - e^x + 1 \quad (4)$$

we have the first and second derivatives to be;

$$f'(x) = 1 - e^x, \quad f''(x) = -e^x \quad (5)$$

at $x = 0$, we have;

$$f(0) = 0, \quad f'(0) = 0 \quad (6)$$

making $p = 0$ a multiple root of multiplicity $m = 2$

To evaluate $N'_f(0) = \lim_{x \rightarrow 0} N'_f(x)$, expand $f(x)$, $f'(x)$, and $f''(x)$ using Taylor series around $x = 0$

$$f(x) = x - \left(1 + x + \frac{x^2}{2} + \mathcal{O}(x^3)\right) + 1 = -\frac{x^2}{2} + \mathcal{O}(x^3) \quad (7)$$

$$f'(x) = 1 - \left(1 + x + \frac{x^2}{2} + \mathcal{O}(x^3)\right) = -x - \frac{x^2}{2} + \mathcal{O}(x^3) \quad (8)$$

$$f''(x) = -1 + \mathcal{O}(x) \quad (9)$$

Substituting these Taylor series into $N'_f(x)$:

$$N'_f(0) = \lim_{x \rightarrow 0} \frac{\left(-\frac{x^2}{2} + \mathcal{O}(x^3)\right)(-1 + \mathcal{O}(x))}{(-x + \mathcal{O}(x^2))^2} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2} + \mathcal{O}(x^3)}{x^2 + \mathcal{O}(x^3)} = \frac{1}{2} \quad (10)$$

Conclusion by the Fixed-Point Theorem, an iteration $p_{n+1} = N_f(p_n)$ converges linearly if $0 < |N'_f(p)| < 1$. Because $N'_f(0) = \frac{1}{2} \neq 0$, Newton's method degrades from quadratic convergence to linear convergence with a rate equal to $N'_f(0) = \frac{1}{2}$.

2. To modify the Newton method for quadratic convergence, we have to consider the multiplicity of the root which is $k = 2$;

We modify the update formula by scaling the step by k :

$$p_{n+1} = p_n - k \frac{f(p_n)}{f'(p_n)} = p_n - 2 \frac{f(p_n)}{f'(p_n)} \quad (11)$$

Let $N_f(x) = x - 2 \frac{f(x)}{f'(x)}$ be the modified iteration function:

$$N'_f(x) = 1 - 2 \frac{[f'(x)]^2 - f(x)f''(x)}{[f'(x)]^2} = -1 + 2 \frac{f(x)f''(x)}{[f'(x)]^2} \quad (12)$$

By Taylor's series expansion of $f(x)$ and taking the limit.

$$N'_f(0) = \lim_{x \rightarrow 0} N'_f(x) = -1 + \lim_{x \rightarrow 0} 2 \frac{\left(-\frac{x^2}{2} + \mathcal{O}(x^3)\right)(-1 + \mathcal{O}(x))}{(-x + \mathcal{O}(x^2))^2} = -1 + \lim_{x \rightarrow 0} 2 \frac{\frac{x^2}{2} + \mathcal{O}(x^3)}{x^2 + \mathcal{O}(x^3)} = -1 + \frac{2}{2} = 0 \quad (13)$$

Because $N'_f(0) = 0$, the modified algorithm restores quadratic convergence of the Newton method. We have the figure below for 6 iterations;

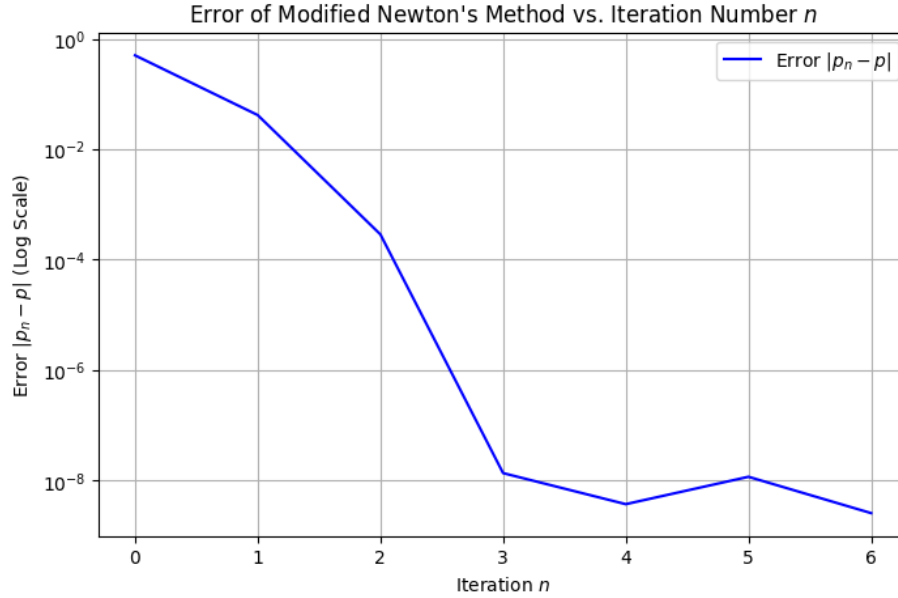


Figure 2: Newton Method

3. Exercise 3

We have that the $\|f - P_{n-1}\|_\infty$ is 0.00090 for the first function and 0.92573 for the second function. We have that the infinity norm error $\|f - P_{n-1}\|_\infty$ of $f(x) = \sin(|x|)$ converges to 0 as $n \rightarrow \infty$. However, the infinity norm error of $f(x) = \text{sign}(x) \cos(x)$ does not converge to 0. It oscillates and remains bounded away from zero. The first function converges mainly because we have uniform convergence to zero guaranteed by the Jackson Theorem for Chebyshev interpolation on compact intervals. On the other hand, the function $f(x) = \text{sign}(x) \cos(x)$ oscillates and fails to converge due to the Gibbs phenomenon. Near $x = 0$, the function function experience wiggling (persistent overshoot and undershoot). The $\|f - P_{n-1}\|_\infty$ never reaches 0

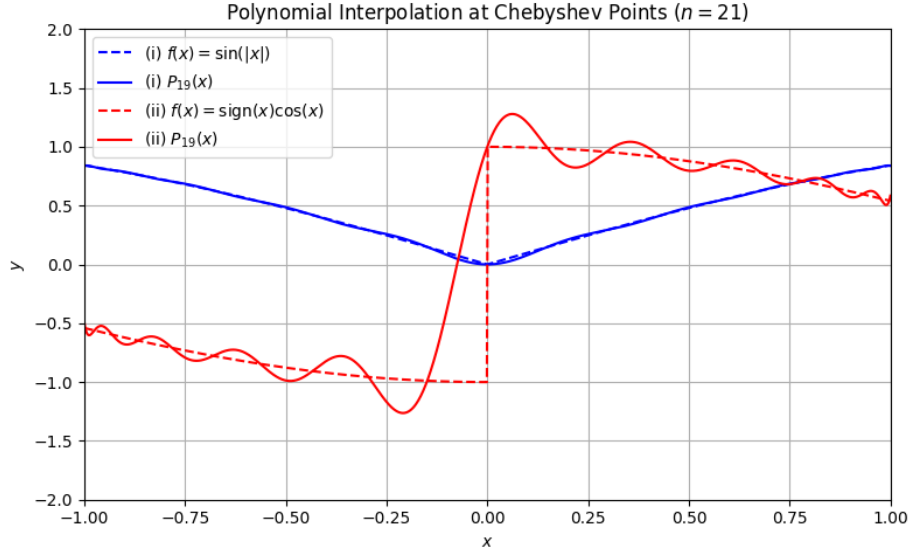


Figure 3: Lagrange Interpolation

4. For the L_2 norm error, both functions converge, however the convergence of the first is faster while the convergence of the second is extremely slow. The function $f_1(x)$ is a continuous function, though not differentiable at $x = 0$, which can be easily approximated by polynomials. The non-smoothness limits its speed of convergence to an algebraic rate. For the other function, we have that the width of the overshoot region shrinks toward 0 as n increases. Because the L^2 norm integrates the squared error over space, the shrinking width allows the integral to approach 0 at a rate of $\mathcal{O}(n^{-1/2})$.

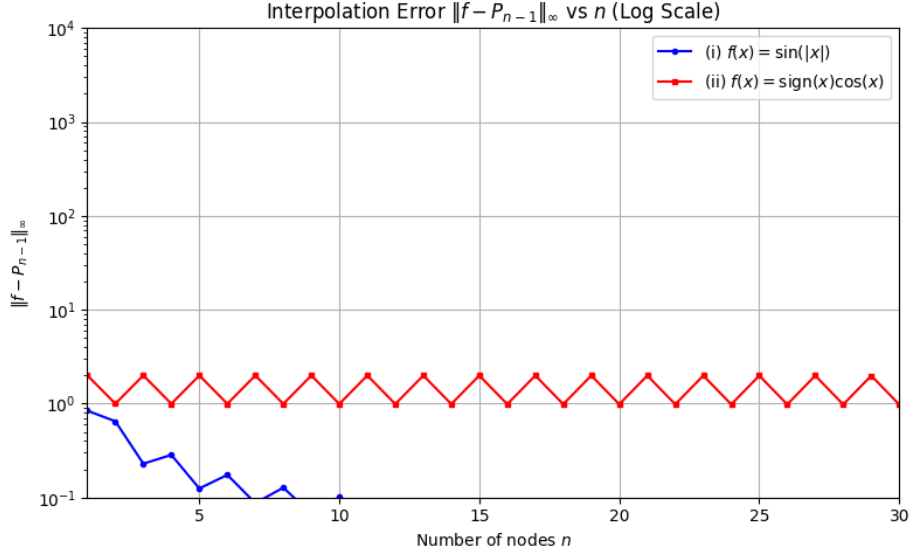


Figure 4: The relative error in the L_{∞} norm

5. We want to approximate $f''(x)$ using the values of $f(x)$ at $x_0 - h$, x_0 and $x_0 + \alpha h$. now suppose we have constant A, B and C such that;

$$f''(x) \approx Af(x_0 - h) + Bf(x_0) + Cf(x_0 + \alpha h) \quad (14)$$

By Taylor series expansion, we have;

$$f(x_0 - h) = f(x_0) - hf'(x_0) + \frac{h^2}{2}f''(x_0) - \frac{h^3}{6}f'''(x_0) + \frac{h^4}{24}f^{(4)}(x_0) - \mathcal{O}(h^5) \quad (15)$$

$$f(x_0 + \alpha h) = f(x_0) + \alpha hf'(x_0) + \frac{\alpha^2 h^2}{2}f''(x_0) + \frac{\alpha^3 h^3}{6}f'''(x_0) + \frac{\alpha^4 h^4}{24}f^{(4)}(x_0) + \mathcal{O}(h^5) \quad (16)$$

We substitute the series above in the approximate, rearrange and group the derivative terms, we have;

$$f''(x_0) \approx (A+B+C)f(x_0) + (-A+\alpha C)hf'(x_0) + (A+\alpha^2 C)\frac{h^2}{2}f''(x_0) + (-A+\alpha^3 C)\frac{h^3}{6}f'''(x_0) + \dots \quad (17)$$

Now comparing terms, we have the following equations;

$$f(x_0) : A + B + C = 0, \quad f'(x_0) : -A + \alpha C = 0 \implies A = \alpha C \quad (18)$$

$$f''(x_0) : (A + \alpha^2 C)\frac{h^2}{2} = 1 \quad (19)$$

solve those equations above we have;

$$A = \frac{2}{(1+\alpha)h^2}, \quad B = -\frac{2}{\alpha h^2}, \quad C = \frac{2}{\alpha(1+\alpha)h^2} \quad (20)$$

The Finite Difference Scheme for $f''(x_0)$ is;

$$f''(x_0) \approx \frac{2}{(1+\alpha)h^2}f(x_0 - h) - \frac{2}{\alpha h^2}f(x_0) + \frac{2}{\alpha(1+\alpha)h^2}f(x_0 + \alpha h) \quad (21)$$

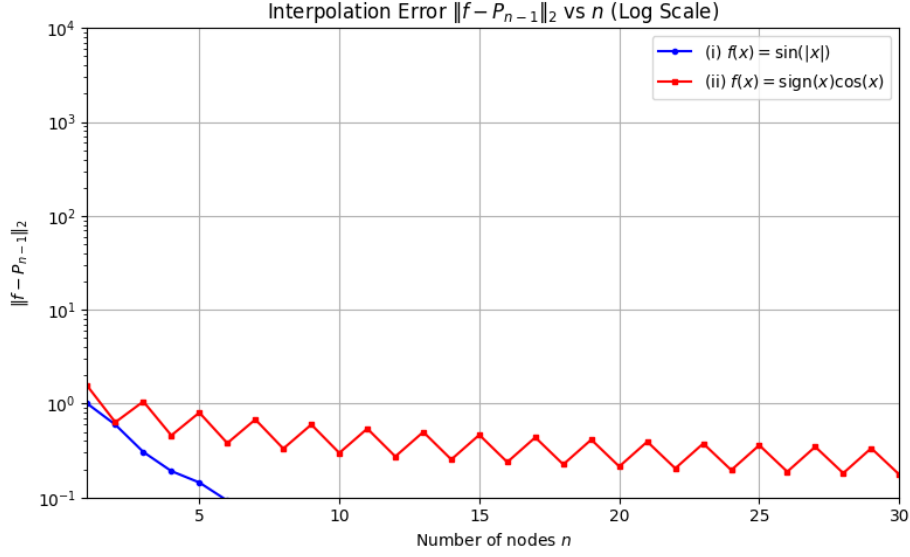


Figure 5: The relative error in the L_2 norm

The error in this approximation is given by;

$$E = f''(x_0) - Af(x_0 - h) - Bf(x_0) - Cf(x_0 + \alpha h) \quad (22)$$

$$E = (A - \alpha^3 C) \frac{h^3}{6} f'''(x_0) - (A + \alpha^4 C) \frac{h^4}{24} f^{(4)}(x_0) + \dots \quad (23)$$

With the values of A and c , we have the error term as;

$$E = -\frac{\alpha - 1}{3} h f'''(x_0) - \frac{1 - \alpha + \alpha^2}{12} h^2 f^{(4)}(x_0) + \mathcal{O}(h^3) \quad (24)$$

6. For $\alpha \neq 1$, the leading error term is $\mathcal{O}(h)$, making the scheme first-order accurate. For $\alpha = 1$ The leading term vanishes ($\alpha - 1 = 0$), reducing the error to $-\frac{h^2}{12} f^{(4)}(x_0) + \mathcal{O}(h^4)$, which gives the standard second-order accurate central difference scheme.